

GA-Driven Hybrid Intelligence for Indonesian Banking Stocks Forecasting: Unifying LSTM, Bi-LSTM, XGBoost, and LightGBM with Statistical Validation and Explainable AI

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Abstract—This paper studies daily closing-price forecasting for major Indonesian banking stocks using a multi-model pipeline with macroeconomic regressors and explainability analysis. The proposed workflow compares four model families (LSTM, BiLSTM, XGBoost, and LightGBM) under two feature settings (price-only and price+macro), with and without genetic algorithm (GA)-based hyperparameter optimization. To improve experimental rigor, we run repeated multi-seed evaluations across multiple banks and report distributional performance using RMSE, MAE, MAPE, and MSE. We further apply statistical significance testing (Wilcoxon signed-rank and Diebold-Mariano) to verify whether observed gains are robust rather than stochastic artifacts. Explainable AI analysis is conducted using SHAP to identify dominant drivers in both local and global perspectives. Initial results indicate that macroeconomic features and GA-based tuning can improve predictive accuracy in selected banks, while gains remain model- and regime-dependent. This paper provides a reproducible baseline for subsequent computational finance studies on Indonesian equities.

Index Terms—stock forecasting, LSTM, BiLSTM, XGBoost, LightGBM, genetic algorithm, explainable AI, SHAP, Diebold-Mariano test

I. INTRODUCTION

Forecasting bank stock prices remains challenging due to non-stationarity, regime shifts, and sensitivity to macroeconomic conditions. In emerging markets such as Indonesia, banking-sector prices are influenced not only by internal price dynamics but also by external drivers including interest rates, inflation, exchange rates, and broad market movements.

Recent work in financial time series forecasting has adopted deep learning models, especially recurrent networks, to capture temporal dependencies. However, many studies still suffer from limited ablation depth, weak statistical validation, and insufficient explainability. These gaps reduce confidence in whether reported improvements generalize across assets and random seeds.

This study develops a reproducible pipeline for Indonesian banking stocks that combines sequence modeling, macroeconomic regressors, repeated-seed evaluation, and formal significance testing. The objective is to provide a strong foundational benchmark that can be extended to more advanced architectures.

The main contributions are as follows:

- A structured ablation design over model type (LSTM vs BiLSTM), feature regime (price-only vs price+macro), and optimization mode (off vs on).
- A multi-bank, multi-seed protocol for robust performance estimation and variance analysis.
- Statistical validation using Wilcoxon signed-rank and Diebold-Mariano tests to assess significance of forecasting improvements.
- Explainability analysis using SHAP to connect model behavior with financially interpretable drivers.

The remainder of this paper is organized as follows. Section II reviews related literature. Section III describes the dataset and preprocessing. Section IV presents the proposed methodology. Section V details the experimental setup. Section VI discusses the results and significance tests. Section VII concludes and outlines future directions.

II. RELATED WORKS

Literature on stock forecasting broadly spans statistical models, machine learning regressors, and deep sequence models. Classical approaches often provide interpretability but struggle with non-linear and non-stationary dynamics. Tree-based learners such as XGBoost and LightGBM improve non-linear modeling and remain strong tabular baselines in financial prediction tasks.

Deep learning approaches, especially LSTM variants, are widely used to model temporal dependencies in market data. Bidirectional recurrent structures can improve representation quality in supervised sequence learning, although performance gains are data-dependent and not universally guaranteed.

Another active direction involves hyperparameter optimization strategies, including random search, Bayesian optimization, and population-based methods. In practice, fair comparison requires matched search budgets and transparent reporting of computational cost.

Explainable AI is increasingly required in computational finance to support model diagnostics and trust. SHAP-based analysis enables both global and local attribution, helping identify which technical or macroeconomic variables drive predictions in different market regimes.

Despite these developments, many studies still lack combined treatment of ablation, repeated-seed robustness,

and formal significance testing. This paper addresses that methodological gap in the context of Indonesian bank stocks.

III. DATASET AND PREPROCESSING

A. Dataset Description

This study focuses on seven Indonesian banking stocks: BBKA, BBNI, BBRI, BBTN, BDMN, BMRI, and BNGA. Daily market data include price and valuation-related variables, while macroeconomic variables include inflation-related indicators, policy rate, exchange rate, and market index features.

The target variable is the daily closing price. The analysis period follows a chronological split to avoid data leakage.

B. Feature Selection

The baseline feature set (price-only regime) includes market micro and technical variables available in the bank-level dataset. The macro-augmented regime extends this set with external variables:

- inflation rate and CPI-derived indicators,
- BI policy rate,
- USD/IDR exchange rate,
- IHSG index level and return proxy.

The final feature schema is fixed per regime before model training.

C. Data Preprocessing

We apply chronological sorting, forward-fill for missing observations, and standard scaling for both features and target during model training. Sequence windows are then constructed with a fixed lookback length.

Time split protocol:

- Training: 2014-01-02 to 2019-12-31
- Validation: 2020-01-02 to 2021-12-31
- Testing: 2022-01-02 to 2024-12-30

This split reflects strict temporal validation and supports fair out-of-sample evaluation.

IV. PROPOSED METHODOLOGY

A. Overall Framework

The pipeline consists of five stages: data loading and scaling, sequence generation, model training, out-of-sample forecasting, and interpretability analysis. We evaluate multiple design factors using controlled ablation to isolate contribution from architecture, macro features, and optimization.

B. Machine Learning Models

1) *LSTM and BiLSTM*: We use a one-layer recurrent encoder followed by a fully connected prediction head. Two architecture variants are compared:

- LSTM: unidirectional temporal encoder.
- BiLSTM: bidirectional temporal encoder with doubled hidden-state output dimension.

2) *Planned Tree-Based Baselines*: To strengthen state-of-the-art comparison, we include XGBoost and LightGBM baselines under matched feature regimes (price-only and price+macro). Together with LSTM and BiLSTM, this forms a four-model benchmark suite.

C. Genetic Algorithm Hyperparameter Optimization

Hyperparameter tuning is formulated as a GA-driven search process with validation loss as the fitness objective. Candidate configurations are evolved across generations using selection, crossover, and mutation to balance exploration and exploitation. The optimization block is applied consistently across comparable model settings under a fixed budget for fair evaluation.

D. Statistical Validation

Beyond average error reporting, we apply:

- Wilcoxon signed-rank test on seed-level paired RMSE,
- Diebold-Mariano test on aligned forecast residuals.

These tests assess whether gains over baseline are statistically significant.

E. Explainable AI using SHAP

We apply SHAP KernelExplainer on selected train/test subsets to estimate feature attribution. Analysis is reported in two forms:

- Global explanation: mean absolute SHAP importance by feature.
- Local explanation: selected instances under volatile periods.

This supports economically meaningful interpretation of model behavior.

V. EXPERIMENTAL SETUP

A. Evaluation Metrics

Primary metrics are MSE, RMSE, MAE, and MAPE on the test partition. We also report seed-wise variability (mean and standard deviation) to characterize stability.

B. Ablation Matrix

The main ablation factors are:

- Model family: LSTM, BiLSTM, XGBoost, and LightGBM
- Features: price-only vs price+macro
- Optimization: GA off vs GA on

This yields a scalable matrix for controlled comparison across deep sequence and tree-based learners.

C. Multi-Seed Protocol

Each configuration is executed with five seeds (42, 52, 62, 72, 82). The complete run table stores configuration meta-data, metrics, and file paths for downstream significance analysis.

D. Implementation Details

Implementation is based on Python and PyTorch. Training uses Adam optimizer, ReduceLROnPlateau scheduling, and early stopping with validation monitoring. Sequence length, forecasting horizon, and tuning budget are controlled via command-line arguments for reproducibility.

E. Cross-Model Fairness Protocol

To maintain fairness, each model family uses comparable data splits, identical seed sets, and matched optimization budgets. This avoids overstating gains from unequal search effort.

TABLE I
PLACEHOLDER FOR MAIN MULTI-SEED PERFORMANCE TABLE (RMSE, MAE, MAPE, MSE).

Item	Description
Rows	Bank $\times$ case
Values	Mean and std across seeds

TABLE II
PLACEHOLDER FOR SIGNIFICANCE TESTS (BEST CASE VS BASELINE).

Item	Description
Tests	Wilcoxon, Diebold-Mariano
Outputs	Test statistic, p-value, improvement (%)

VI. RESULTS AND DISCUSSION

A. Performance Comparison Across Configurations

Table I will summarize per-bank performance across ablation cases using mean $\pm$ std over seeds. Initial analysis should highlight: (i) whether macro features improve accuracy, (ii) whether BiLSTM consistently outperforms LSTM, and (iii) whether optimization yields robust gains.

B. Statistical Significance Analysis

Table II will report Wilcoxon and Diebold-Mariano statistics for best-case vs baseline comparisons per bank. This section should distinguish between numerical improvement and statistically significant improvement.

C. Explainability Analysis

Global SHAP plots will identify dominant features under each regime. Local SHAP examples should be selected from volatile market intervals to explain prediction behavior under stress. Discussion will emphasize financial interpretation rather than only ranking statistics.

D. Discussion and Threats to Validity

Key validity points include data horizon selection, sensitivity to split boundaries, and potential overfitting risk under repeated tuning. Future iterations should include walk-forward validation and external baseline models (XGBoost/LightGBM) under matched budgets.

VII. CONCLUSION

This paper presents a reproducible framework for forecasting Indonesian bank stock prices using LSTM-family models, macroeconomic features, multi-seed evaluation, statistical significance testing, and SHAP-based explainability. The framework is designed as a foundational benchmark for computational finance research in the Indonesian market.

The current evidence indicates that architecture choice, macroeconomic context, and optimization strategy each contribute differently across banks, motivating bank-specific model selection rather than one-size-fits-all deployment.

Immediate future work includes: (i) expanding and stress-testing the four-model suite (LSTM, BiLSTM, XGBoost, LightGBM) with matched GA budgets, (ii) extending validation to walk-forward protocols, and (iii) adding architecture upgrades such as attention-based variants.